PLATE NO. VIII · TRANSLATION GEOMETRY

SPECIMEN: DPT-2028 · ARTICULATED KINEMATICS

DATUM: 40°01'14"N 75°09'22"W [GLENSIDE TERRESTRIAL]

INSTRUMENTATION: 7-DOF SERIAL MANIFOLD × RIEMANNIAN PHASE SPACE

METRIC TENSOR: $g_{ij} = J_i^T(q)\, M(q)\, J_j(q)$ · INVARIANT TORSION

KINEMATIC MONISM

THE SPATIAL GEOMETRY OF ARTICULATED MANIFOLDS

A HARMONIC SYNTHESIS OF HUMAN BIOMECHANICS & INDUSTRIAL ROBOTIC DEGREES OF FREEDOM

SYSTEM METRIC: SO(3) × SE(3)

DEGREE OF FREEDOM: 07 [$q_1 .. q_7$]

ANGULAR ENVELOPE: 0.00° → 135.24°

VELOCITY VECTOR: $\dot{q}(t) \in C^2[0, T]$

STATUS: VERIFIED MATHEMATICAL STUDY

FIG. 01 · SAGITTAL PROJECTION

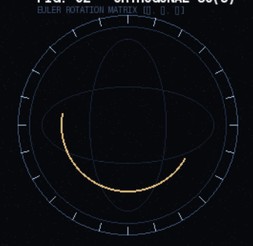
TIBIOFEMORAL RANGE: 0°–135°



ICR OFFSET: +3.2mm

FIG. 02 · ORTHOGONAL SO(3)

EULER ROTATION MATRIX [α · β · γ]



DET(R) = +1.00000



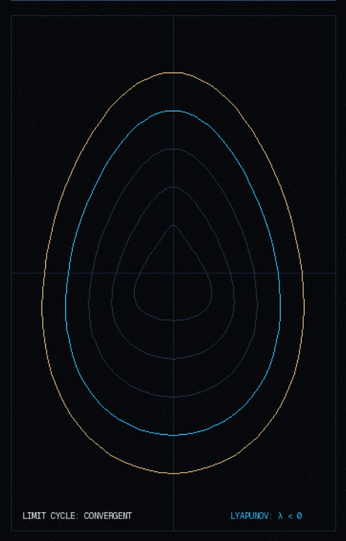
DENAVIT-HARTENBERG COORDINATE CHAIN

KINEMATIC LINK TRANSFORMATION MATRICES

LINK 01	θ_1 = -045.0°	θ: 0° | d: 0mm
LINK 02	θ_2 = -016.5°	θ: 90° | d: 45mm
LINK 03	θ_3 = +012.0°	θ: 0° | d: 90mm
LINK 04	θ_4 = +040.5°	θ: 90° | d: 135mm
LINK 05	θ_5 = +069.0°	θ: 0° | d: 180mm
LINK 06	θ_6 = +097.5°	θ: 90° | d: 225mm
LINK 07	θ_7 = +126.0°	θ: 0° | d: 270mm

FIG. 03 · PHASE SPACE ATTRACTOR

ANGULAR DISPLACEMENT vs. VELOCITY [θ vs. θ̇]



LIMIT CYCLE: CONVERGENT

LYAPUNOV: λ < 0

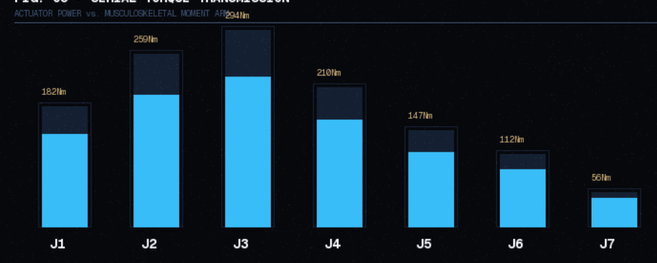
FIG. 04 · GROUND REACTION FORCE HARMONICS

FOURIER SPECTRAL DECOMPOSITION OF STRIDE CADENCE



FIG. 05 · SERIAL TORQUE TRANSMISSION

ACTUATOR POWER vs. MUSCULOSKELETAL MOMENT ARM



[AXIOM 01 · RIEMANNIAN KINEMATICS]

Space and form do not merely contain movement;
they are the crystallized residual trace of motion itself.
The boundary between mechanical serial chains and biological limbs
dissolves into continuous vector manifolds and invariant coordinate strata.

$$d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = \tau - J_c^T(q)\cdot\lambda \quad | \quad \nabla H(p,\ q) = 0$$

LEIBNIZ DIFFERENTIAL CONTINUUM · ARC LENGTH $S = \int ||r'(t)||\, dt$

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[AXIOM 02 · KINETIC EQUILIBRIUM]

Every articulation obeys an overarching harmonic law.
In biological structures and artificial manipulators alike,
force and compliance achieve resolution without decorative excess.
Information lives in design, precision, and spatial cadence.

$$T_7^0(q) = \prod A_i(q_i) \quad | \quad \text{rank}(J(q)) = 6 \quad | \quad \det(M) > 0$$

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